

Lucien Brey

MACHINE LEARNING CHEMIST

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Working experience

Atinary Technologies Inc.

Lausanne, Switzerland

MACHINE LEARNING SCIENTIST

Nov. 2024 - Now

- Development and maintenance of proprietary Bayesian optimization algorithms
- Augment optimization scheme with physical and chemical prior
- Generative modeling for high combinatorial space optimization
- Collaboration with academic partners such as Prof. J.M Hernandez-Lobato or Prof. Philippe Schwaller

Atinary Technologies Inc.

Lausanne, Switzerland

MACHINE LEARNING AND DATA SCIENCE

Aug. 2023 - Jan. 2024

INTERN

- Developing a NLP-based tool to enable database and literature search from user request for more efficient transfer learning
- Benchmarking and optimization of existing algorithms

EPFL

Lausanne, Switzerland

TEACHING ASSISTANT

- TA in the Advanced General Chemistry course for first year bachelor students in EPFL Life Sciences program and UNIL Geosciences — 2020-2023

Projects

MIT, Prof. Rafa Gómez Bombarelli

Cambridge MA, USA

VISITING RESEARCHER

Mar. 2024 - Sep. 2024

- Master thesis on "Differentiable inverse design of metal oxides and perovskites catalyst via fine-tuning of surface oxygens site-properties"
- Built gradient-based optimization pipeline for structural inverse design of catalysts

EPFL, Prof. Philippe Schwaller

Lausanne, Switzerland

PROJECT STUDENT

Feb. 2023 - Jul. 2023

- Completing USPTO-MIT database with stoichiometry and reaction conditions, together with by-products, based on reference private database
- Using transformers for reaction completion and byproduct prediction
- Developed an Streamlit web app for reaction prediction

EPFL, Prof. Clémence Corminboeuf

Lausanne, Switzerland

PROJECT STUDENT

Sep. 2022 - Jan. 2023

- Building database based on experimental results for enantioselectivity of catalyst/substrate couples for Pictet-Spengler reaction
- Developing quantum machine learning model to shape a good fitness function capable to predict the key parameter in selectivity for a new catalyst/substrate structure

KTH, Prof. Mårten Ahlquist

Stockholm, Sweden

RESEARCHER ASSISTANT

Jan 2022-Jun.2022

- QM-based mechanistic rationalization of ligand-controlled product selectivity in electrochemical carbon dioxide reduction using Mg-bipyridine catalysts
- Structure optimization, transition state search and 3D modelling on Schrodinger Maestro software

Education

École Polytechnique

Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

STUDENT

2019-2024

- Master in Molecular and Biological Chemistry — Sep. 2022-Sep. 2024
- Bachelor in Chemistry and Chemical Engineering — Sep.2019-June 2022

Kungliga Tekniska

Högskolan (KTH)

Stockholm, Sweden

EXCHANGE YEAR

Aug. 2021 - June 2022

- Exchange year abroad, discovering the culture, language and educational system in Sweden

Skills

Languages

French C2 | English C2 | Spanish C1 | Italian A2 | German A1

Programming (Python) | Bayesian Optimization | Gaussian Processes | Reinforcement Learning |

Skills

Organic Chemistry | Inorganic Chemistry | Quantum Chemistry | Quantum Molecular Dynamics | Photochemistry | Analytical Chemistry

Packages

Torch | BoTorch | GPyTorch | Numpy | SKLearn

Associations

EPFL Chemistry Student Society (AdEC)

Lausanne, Switzerland

PRESIDENT

Sep. 2022-Aug. 2023

- Organizing events such as Weekend Ski, Fondue de Section, Welcome Day for new students, quiznights in a bar etc.
- Coordination of seminars with the Swiss Chemical Society (SCS), professors, companies and alumni

KTH THS International

Stockholm, Sweden

BUDDY

Jan. 2022-Mar.2022

- Responsible of a group of international students, organizing, preparing and staffing during events such as museum visits or trivia nights

Coaching Chimie EPFL

Lausanne, Switzerland

TREASURER

Sep. 2020-Jun. 2021

- In charge of the association expenses, expense report, budget etc.
- Organization of events (game nights, picnics...) for the first year students to help them socialize and get introduced to the student life

Publications

S. Gallarati, P. van Gerwen, R. Laplaza, L. Brey, A.

Makaveev and C. Corminboeuf

A genetic optimization strategy with generality in asymmetric organocatalysis as a primary target,

Chemical Science, Vol. 15, Issue 10, 3640-3660

2024